TRIBHUVAN UNIVERSITY

INSTITUTE OF ENGINEERING

Examination Control Division

2075 Chaitra

Exam.	**Regular / Back**		
Level	BE	**Full Marks**	80
Programme	BCE	**Pass Marks**	32
Year / Part	IV / I	**Time**	3 hrs.

Subject: - Seismic Resistant Design of Masonry Structure *(Elective I) (CE72502)*

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ *Attempt **All** questions.*
- ✓ *The figures in the margin indicate **Full Marks**.*
- ✓ *Assume suitable data if necessary.*
- ✓ ***IS-1905, IS-1893, NBC are allowed.***

1. Discuss the design criteria and design steps in the design of masonry shear walls (10)

2. Explain different modes of failure in Brick Masonry structures with sketches for both static and earthquake load (10)

3. Explain following and state the influence of it on strength of the wall: (12)
   i) Slenderness ratio.
   ii) Shape modification factor

4. Design interior wall of two storeyed building to supports 3 m. wide slab of 100 mm. thick. The height of the wall is 3.5 m. and is unstiffened. The roof and floor loadings are as follows; (18)
   Live load on roof = 2 kN/sq. m
   Live load on floor = 3 kN/sq. m .
   Weight of slab = 2.5 kN/sq. m
   Weight of floor finish = 1 Kn/sq. m

5. Design a reinforced brick masonry lintel over a window opening of span 1.5 m. The thickness of wall is 230mm and the height of the brick above lintel to ceiling is 90cm. and length of the wall on either side of the lintel is 1.5m. Use brick masonry of characteristic strength 10 N/sq. mm and fy 415 steel bar. (18)

6. Determine equivalent eccentricity and eccentricity ratio of a 230 mm thick and 3 m high wall. The wall is subjected an axial load of 20 kN/m and an eccentric load of 12 kN/m at distance of 6 mm from the center of wall. (12)

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2076 Chaitra

Exam.	Regular		
Level	BE	Full Marks	80
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Year / Part	IV / I	Time	3 hrs.

Subject: - Seismic Resistant Design of Masonry Structures *(Elective I) (CE72502)*

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ *Attempt **All** questions.*
- ✓ *The figures in the margin indicate **Full Marks**.*
- ✓ **NBC 105, IS 1893, IS 1905 codes are allowed.**
- ✓ *Assume suitable data if necessary.*

1. a) State and explain briefly the factors affecting the compressive strength of brick masonry. [8]

   b) Design a lintel bands for a single room one storey building of 4m breadth, 6m length and 3m ceiling height. The thickness of brick masonry wall are 220mm and height of the lintel from floor is 2.1m. The roof is of RCC flat slab and design earthquake coefficient is 0.36. Neglect the opening on walls. Show the design details in sketch. [12]

2. a) Explain the design criteria of walls subjected to transverse loading. [8]

   b) Design a reinforced brick masonary lintel subjected to triangular loading for a window opening of span 1.6m. The thickness of the wall is 220mm and height of the brickwork above the lintel is 0.9m. Length of the wall on either side of the lintel is more than half of the span of the lintel. [12]

3. a) Discuss in brief failure mechanism of masonry structures and remedial measures. [10]

   b) Design interior wall of two storeyed building to carry 120mm thick RCC floor and roof slab with 3m ceiling height. The wall is unstiffened and its supports 4m wide slabs. Take live load on roof and floor 2kN/sq.m. [10]

4. a) Determine the seismic coefficient and storey seismic forces for a single room double storey health post building of 4m breadth, 6m length and 3m ceiling height loacated at Kathmandu. The thickness of brick masonary wall are 220mm and height of the lintel from floor is 2.1m. The floor and roof is of RCC flat slab of 15cm thick. Neglect the opening on walls and assume foundation on hard soil. [12]

   b) Explain following applied to masonry structrues. [4×2]

      (i) Stress reduction factor
      (ii) Confined masonry

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2079 Bhadra

Exam.		Regular	
Level	BE	Full Marks	80
Programme	BCE	Pass Marks	32
Year / Part	IV / I	Time	3 hrs.

Subject: - Seismic Resistant Design of Masonry Structure (*Elective I*)(*CE72502*)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ *Attempt **All** questions.*
- ✓ *The figures in the margin indicate **Full Marks**.*
- ✓ ***IS: 1905-1987 codes are allowed.***
- ✓ *Assume suitable data if necessary.*

1. a) Define elastic rebound theory. Explain the origin of earthquakes in Himalayas. [8]

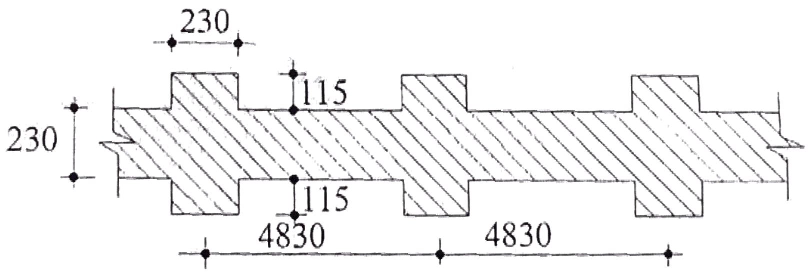
   b) Explain the magnitude and intensity of the earthquake. Compare the MMI scale with the Richter scale. [8]

2. a) Explain load-bearing masonry walls, infill masonry, and confined masonry. [8]

   b) Differentiate between in-plane and out-of plane behaviour of masonry structures. Explain the differentiate failure modes of masonry structures with a neat sketch. [8]

3. a) Define repair, restoration, and retrofitting. [4]

   b) Design an interior wall of a single-storeyed workshop of height 5.4 m supporting an RCC roof. The bottom of the wall rests over a foundation block. Assume roof load equal to 45 kN/m. [12]



4. a) What are the steps involved in retrofitting strategies? [4]

   b) A wall 20 cm thick, using modular bricks carries at the top a load of 80 kN/m having a resultant eccentricity ratio of e = 1/12. Wall is 5 m long between cross walls and is of 3.4 m clear height between RCC slabs at top and bottom. What should be the strength of brick and the grade of mortar? [12]

5. a) What is reinforced masonry? Explain. [4]

   b) Design a shear wall 4.2 m long and 3.3 m high to resist a horizontal earthquake force in its plane. The wall is tied with metal anchors at the top and bottom supports. Assume the seismic load to be uniformly distributed across the height of the wall. Earthquake acceleration = 0.1 g. [12]

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2078 Bhadra

Exam.		**Regular**	
Level	BE	**Full Marks**	80
Programme	BCE	**Pass Marks**	32
Year / Part	IV / I	**Time**	3 hrs.

Subject: - Seismic Resistant Design of Masonry Structures *(Elective I)(CE 72502)*

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ *Attempt **All** questions.*
- ✓ *The figures in the margin indicate **Full Marks**.*
- ✓ ***Necessary Codes are allowed to use.***
- ✓ *Assume suitable data if necessary.*

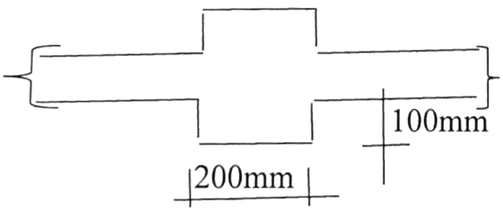
1. a) Explain Richter scale and MMI SCALE. [6]

   b) Describe in brief with sketch major Failure mechanism of masonry building during Gorkha earthquake. [6]

2. a) Define masonry structure type based on materials with its advantages and disadvantages. [6]

   b) Describe affecting factors and relation between unit and mortar on strength of brick masonry. [6]

3. Design an interior solid wall with pier of a single storey structure supporting a RCC roof of height 6m. The pier of dimension shown in figure are located at 4 m C/C along the wall. Take load from roof equal to 50 KN /m. [14]



Pier at 4 m c/c

4. Design an Interior cavity wall of a two storeyed building with ceiling height of 3 m. The wall is unstiffened and 4 m long. Take load from roof 10 kN/m and load from floor 15 kN/m. [14]

5. Design an exterior wall of a single storey workshop of height 4 m. Take load on wall as follows; 30 kN/m from the roof and wind load 750 N/m^2 . The wall is connected by metal anchor with roof structure. [14]

6. Design a shear wall of 5 m long and 4 m high to resist a horizontal earthquake force in its plane under seismic loading . Take seismic load 0.3g acceleration uniformly distributed across the entire height of wall. The wall is connected with metal anchors at the top and bottom supports. Neglect the load from floor. [14]

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